

Relation Algebra

§15. Dynamic Programming

§15.1. Example of Hylo $(T)^\circ(h)$: segmenting a list

- $(T) = \text{concat}$ (15.1a)
 $T = [\text{nil}, \text{cat}]$ takes a non-empty first list, so T° cuts a non-empty prefix off a list every way, and folding with T flattens a list of segments
- $(h) = \text{list}(\text{sum})$
 $h = [\text{nil}, \text{cons}(\text{sum} \times 1)]$ puts a segment's sum in front, so folding with h sums every segment
- $H = \text{partition list}(\text{sum})$
 H segments a list every way, then sums each segment; with R comparing the largest entry, M cuts the list so that the largest segment sum is as small as possible

(15.1b)

x	h(x)
$\text{inl}(())$	$[\]$
$\text{inr}([c], [])$	$[c]$
$\text{inr}([a, b], [c])$	$[a + b, c]$
$\text{inr}([a], [b + c])$	$[a, b + c]$

- $H([a, b, c], \text{ys}) \iff \text{ys} = [a + b + c] \vee \text{ys} = [a, b + c] \vee \text{ys} = [a + b, c] \vee \text{ys} = [a, b, c]$ (15.1c)
 H sends $[a, b, c]$ to the sums of its four segmentations and nothing else
- $T^\circ F(H) h = H$
cut once, solve the rest, put the pieces back together

§15.2. Theory

(15.2a)

$h : FB \rightarrow B$ a map, $T : FA \rightarrow A$ an F -algebra, $R : B \rightarrow B$.

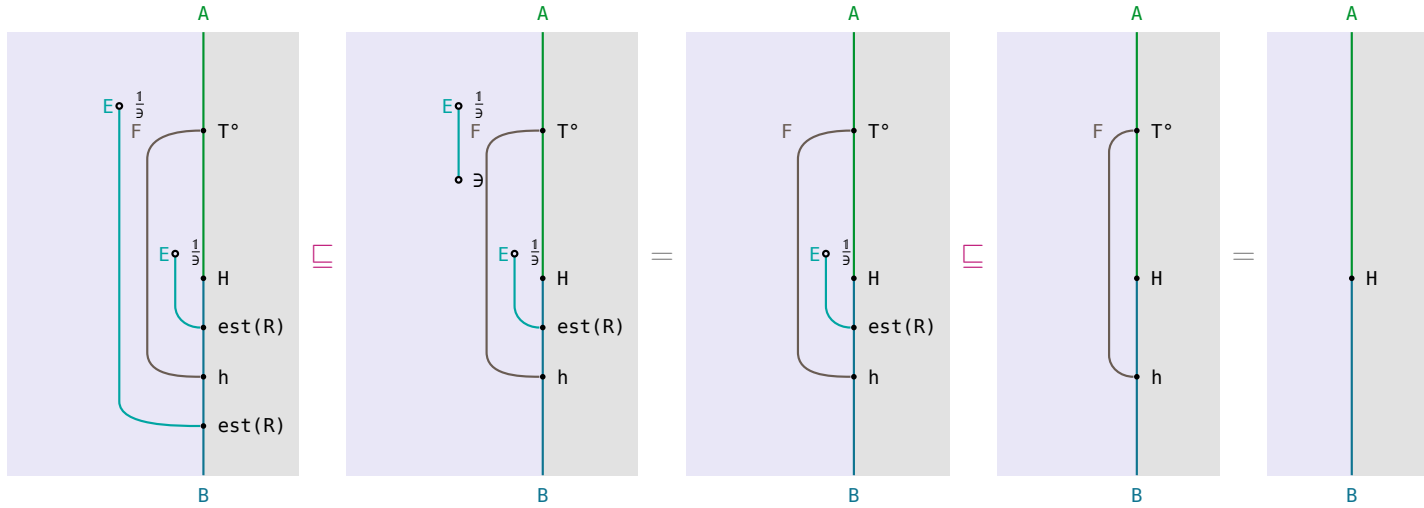
$H \triangleq (T)^\circ(h) : A \rightarrow B$, $M \triangleq \frac{H}{\exists} \text{est}(R)$ the problem to be solved, $(\mu X : G(X))$ as in $\triangleq \mu$.

PreservesRecip(F) $\wedge$ Map(h) $\wedge$ MonotonicAlg(h, R$^\circ$) $\wedge$ R$^\circ$R$^\circ \sqsubseteq$ R$^\circ \implies (\mu X : \frac{T^\circ}{\exists} P(F(X)h) \text{est}(R)) \sqsubseteq \frac{H}{\exists} \text{est}(R)$ (15.2b)
every answer the recursion returns is an optimal one • $\frac{T^\circ}{\exists}$ takes the input apart one step every way; $F(X)$ solves each part by the recursion X; h assembles each candidate; $\text{est}(R)$ keeps a best one • $\sqsubseteq M$: that never misses the optimum, though all candidates are never listed as H would • h monotonic on R: a better part never makes the whole worse • $\sqsubseteq$, not $=$: some optima may not be returned; that the recursion returns anything at all needs T° to stop taking apart — dynamic programming theorem

$$T^\circ F(H)h=H \Rightarrow \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R))h)\text{est}(R)\sqsubseteq H$$

(15.2c)

taking the input apart every way T allows ($\frac{T^\circ}{\exists}$), solving each part by M and keeping an optimum ($P(F(M)h)$
 $\text{est}(R)$) returns only what H returns



$$\begin{aligned} & A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{P(F(\frac{H}{\exists} \text{est}(R))h)} EB \xrightarrow{\text{est}(R)} B \\ \sqsubseteq & A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\exists} FA \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B \\ = & A \xrightarrow{T} FA \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B \\ \sqsubseteq & A \xrightarrow{T} FA \xrightarrow{F(H)} FB \xrightarrow{h} B \\ = & A \xrightarrow{H} B \end{aligned}$$

$$(9.4) P(X)\text{est}(R)\sqsubseteq \exists X \text{ at } X \triangleq F(\frac{H}{\exists} \text{est}(R))h \text{ — est-710}$$

$$\frac{T^\circ}{\exists} \exists = T^\circ \text{ — pow-laws}$$

$$M \triangleq \frac{H}{\exists} \text{est}(R) \sqsubseteq \frac{H}{\exists} \exists = H \text{ — est-up}$$

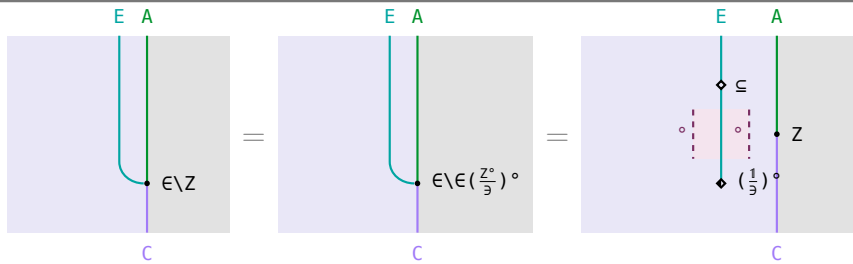
$$T^\circ F(H)h=H: H \triangleq (T)^\circ (h) \text{ and hylo-fix}$$

§15.2.1. $\epsilon \setminus$ as a composite

$$\epsilon \setminus Z = \epsilon (\frac{Z^\circ}{\exists})^\circ$$

(15.2.1a)

xs is related to c by $\epsilon \setminus Z$ exactly when $xs \subseteq \Lambda(Z^\circ)(c)$, the set of every a with $a Z c$



$$\begin{aligned} & EA \xrightarrow{\epsilon \setminus Z} C \\ = & EA \xrightarrow{\epsilon \setminus \epsilon (\frac{Z^\circ}{\exists})^\circ} C \\ = & EA \xrightarrow{\subseteq} E(Z^\circ) \xrightarrow{EC} \frac{1}{\exists} \xrightarrow{\quad} C \end{aligned}$$

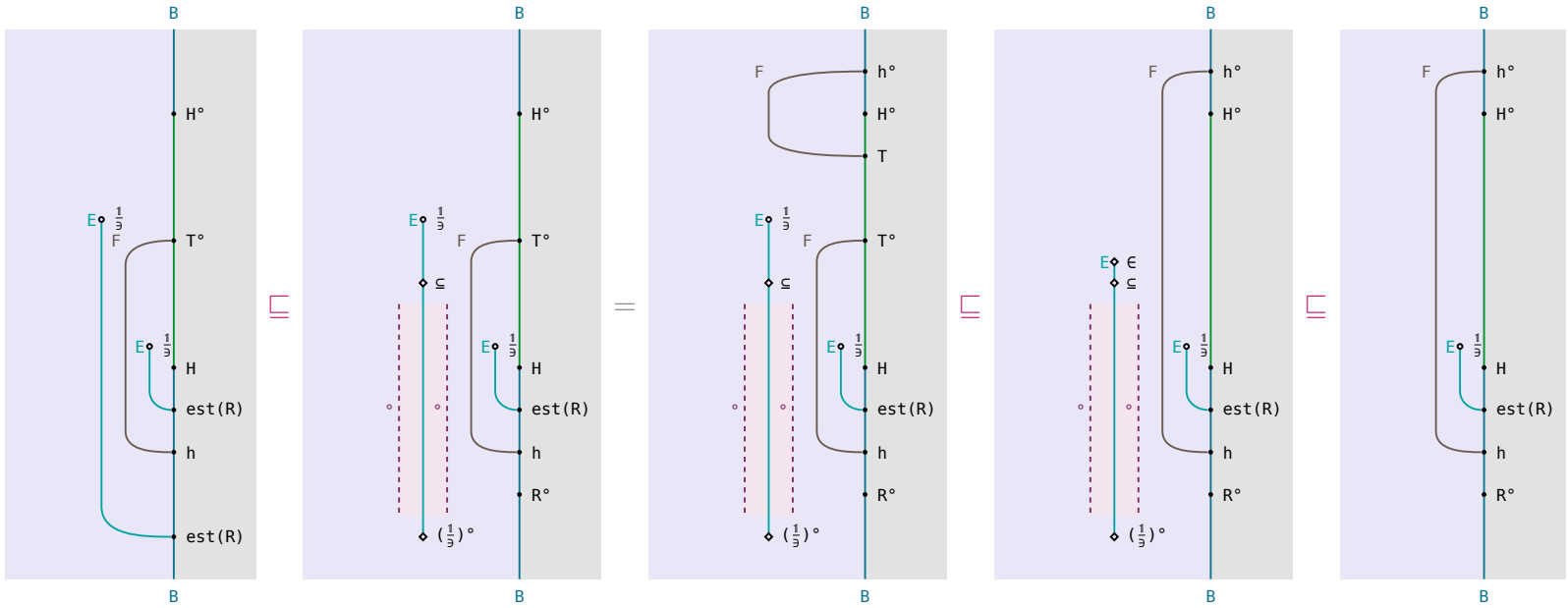
$$Z = \epsilon \Lambda(Z^\circ)^\circ: \text{the converse of } \Lambda(Z^\circ) \exists = Z^\circ \text{ — pow-laws}$$

$$X \setminus (Yf^\circ) = (X \setminus Y) f^\circ \text{ for a map } f \text{ (not a tabulated row), at the map } f \triangleq \Lambda(Z^\circ); \subseteq \triangleq \epsilon \setminus \epsilon$$

$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonotonicAlg}(h, R^\circ) \wedge R^\circ R^\circ \subseteq R^\circ \wedge T^\circ F(H)h = H \implies H^\circ \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R))h) \text{est}(R) \subseteq R^\circ \quad (15.2.1b)$$

$$\text{est}(R))h) \text{est}(R) \subseteq R^\circ$$

for every b that H returns from an input, the step $\frac{T^\circ}{\exists} P(F(M)h) \text{est}(R)$ returns from that input only b' with R relating b' to b



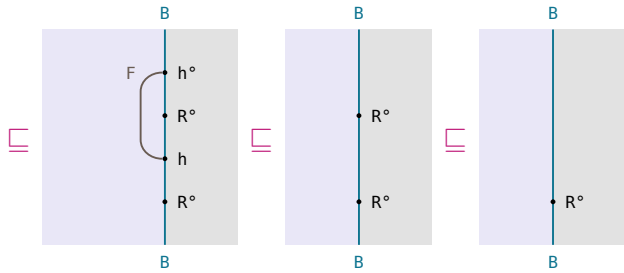
$$\begin{aligned} & B \xrightarrow{H} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{P(F(\frac{H}{\exists} \text{est}(R))h)} EB \xrightarrow{\text{est}(R)} B \\ \subseteq & B \xrightarrow{H} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\epsilon \setminus F(\frac{H}{\exists} \text{est}(R))h R^\circ} B \\ = & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{T} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\epsilon \setminus F(\frac{H}{\exists} \text{est}(R))h R^\circ} B \\ \subseteq & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{\epsilon} EFA \xrightarrow{\epsilon \setminus F(\frac{H}{\exists} \text{est}(R))h R^\circ} B \\ \subseteq & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} R \xrightarrow{} B \end{aligned}$$

$$(9.4) P(X) \text{est}(R) \subseteq \epsilon \setminus (XR^\circ) \text{ at } X \triangleq F(\frac{H}{\exists} \text{est}(R))h \text{ — est-710}$$

$$H^\circ = h^\circ F(H^\circ)T: H \triangleq (T)^\circ(h) \text{ and hydro-fix}$$

$$T \frac{T^\circ}{\exists} \subseteq \epsilon: \frac{T^\circ}{\exists} \text{ a map, } \frac{T^\circ}{\exists} \exists = T^\circ \text{ — pow-laws}$$

$$\epsilon(\epsilon \setminus Y) \subseteq Y \text{ at } Y \triangleq F(M)h R^\circ \text{ — division}$$



$$\begin{aligned} \subseteq & B \xrightarrow{h} F(R^\circ) \xrightarrow{FB} h \xrightarrow{R} B \\ \subseteq & B \xrightarrow{R} R \xrightarrow{} B \\ \subseteq & B \xrightarrow{R} B \end{aligned}$$

$$H^\circ M \subseteq R^\circ: M \triangleq \frac{H}{\exists} \text{est}(R) \text{ — est-up}$$

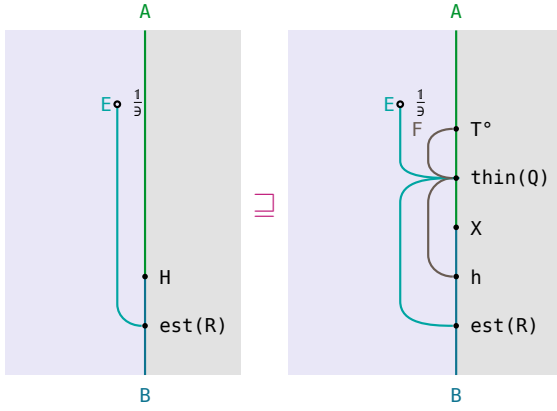
$$h^\circ F(R^\circ)h \subseteq R^\circ: h \text{ monotonic on } R^\circ$$

$$R^\circ R^\circ \subseteq R^\circ: R \text{ transitive}$$

$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonotonicAlg}(h, R^\circ) \wedge R^\circ R^\circ \subseteq R^\circ \wedge QF(H) h \subseteq F(H) h R \implies (\mu X : \frac{T^\circ}{\exists} \quad (15.2.1c)$$

$$\text{thin}(Q) P(F(X)h) \text{est}(R) \subseteq \frac{H}{\exists} \text{est}(R)$$

an optimum over everything H returns is reached by taking the input apart every way T allows, dropping the parts that can never win, solving each of the rest and keeping one optimum



$$A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(H)} EB \xrightarrow{\text{est}(R)} B$$

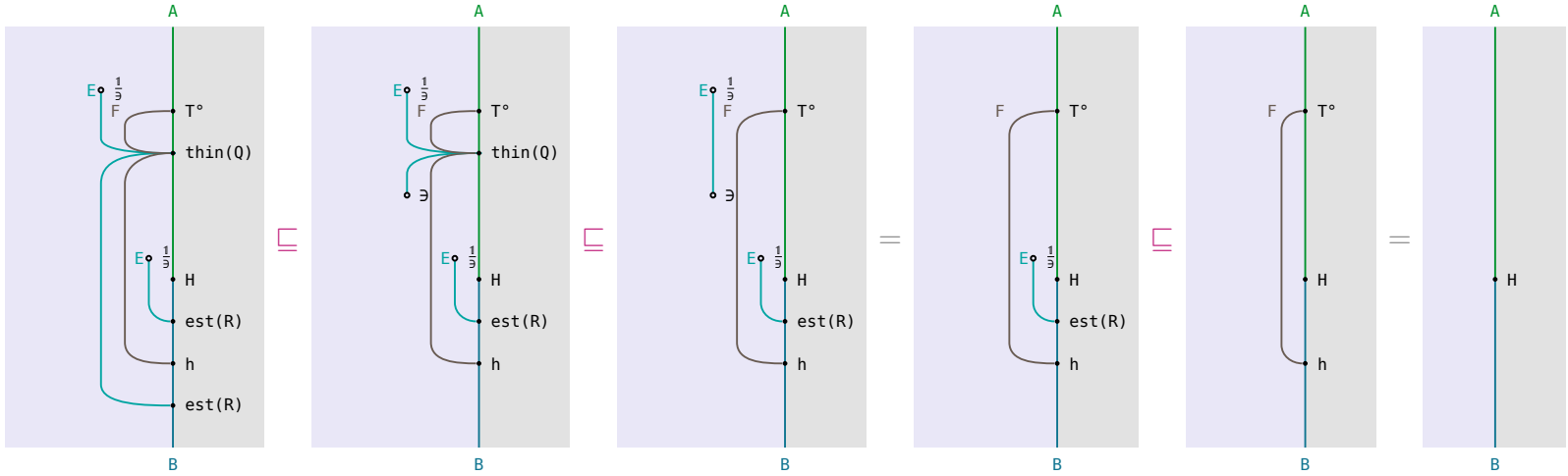
the problem to be solved, $H \triangleq (T)^\circ(h) \triangleq H$

$$A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} \text{thin}(Q) \xrightarrow{EFA} P(F(X)h) \xrightarrow{EB} \text{est}(R) \xrightarrow{} B$$

$\sqsubseteq$ h monotonic on R and Q a preorder with $QF(H)h \subseteq F(H)hR$; $\text{thin}(Q)$ as in thin-laws. Theorem 9.1 is this with the thinning step dropped — $1 \subseteq \text{thin}(Q)$. Knaster–Tarski leaves (9.1): the body at M is $\subseteq M$; $M = H \cap (H^\circ \setminus R^\circ)$ splits that into (9.2) and (9.3) below — est-up. The fixed point is unique and entire when T° followed by F 's membership relation is inductive, $\frac{T^\circ}{\exists}$ finite and non-empty, R connected

$$T^\circ F(H)h = H \implies \frac{T^\circ}{\exists} \text{thin}(Q) P(F(\frac{H}{\exists} \text{est}(R))h) \text{est}(R) \subseteq H \quad (15.2.1d)$$

taking the input apart every way T allows ($\frac{T^\circ}{\exists}$), dropping the parts $\text{thin}(Q)$ rejects, solving each rest by M and keeping an optimum returns only what H returns



$$A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} \text{thin}(Q) \xrightarrow{EFA} P(F(\frac{H}{\exists} \text{est}(R))h) \xrightarrow{EB} \text{est}(R) \xrightarrow{} B \quad (9.2)$$

$$\sqsubseteq A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} \text{thin}(Q) \xrightarrow{EFA} \exists \text{FA} \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B \quad (9.4) \quad P(X) \text{est}(R) \subseteq \exists X \text{ at } X \triangleq F(\frac{H}{\exists} \text{est}(R))h \text{ — est-710}$$

$$\sqsubseteq A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\exists} FA \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B$$

$\text{thin}(Q) \exists \exists$ — thin-laws

$$= A \xrightarrow{T} FA \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B$$

$\frac{T^\circ}{\exists} \exists = T^\circ$ — pow-laws

$$\sqsubseteq A \xrightarrow{T} FA \xrightarrow{F(H)} FB \xrightarrow{h} B$$

$M \subseteq \frac{H}{\exists} \exists = H$ — est-up

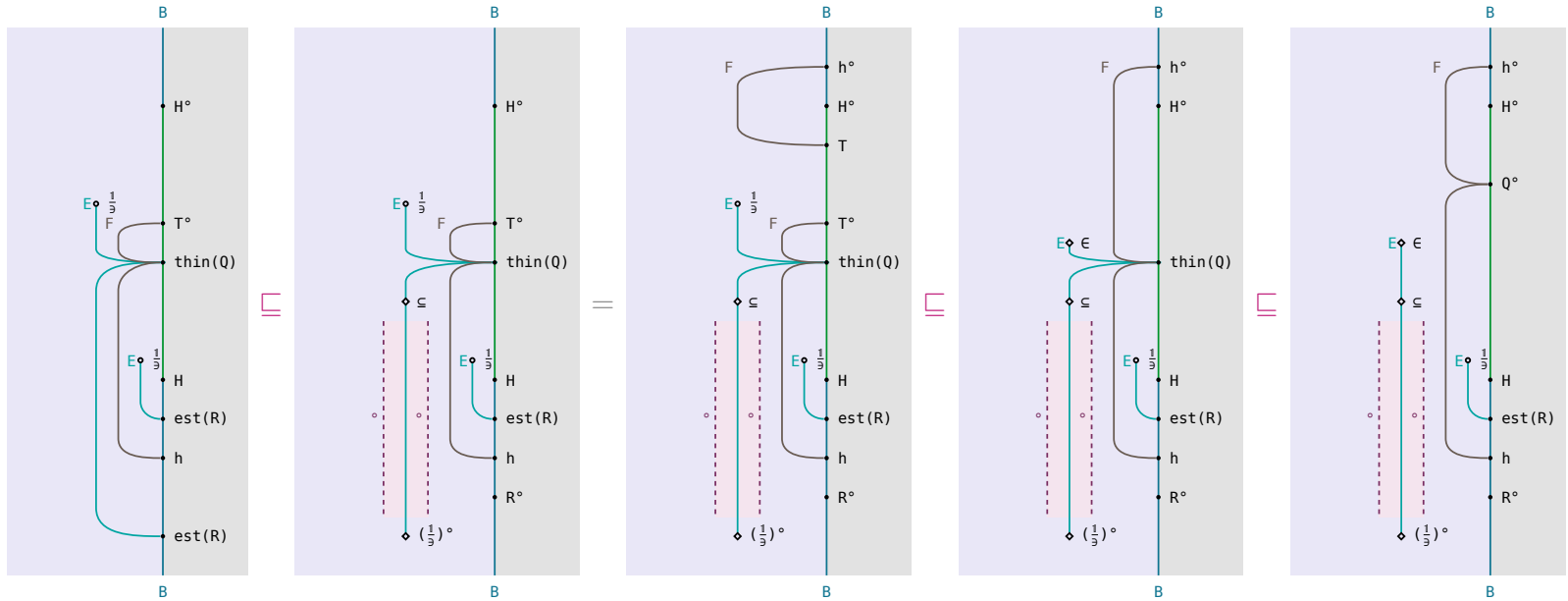
$$= A \xrightarrow{H} B$$

$T^\circ F(H)h = H$: $H \triangleq (T)^\circ(h)$ and hylo-fix

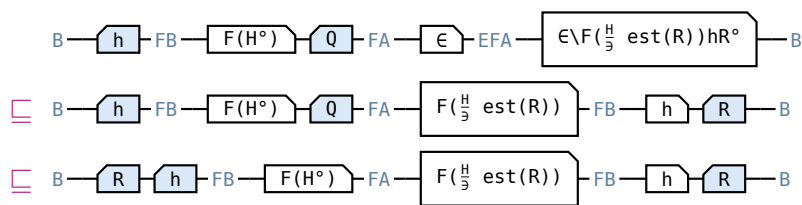
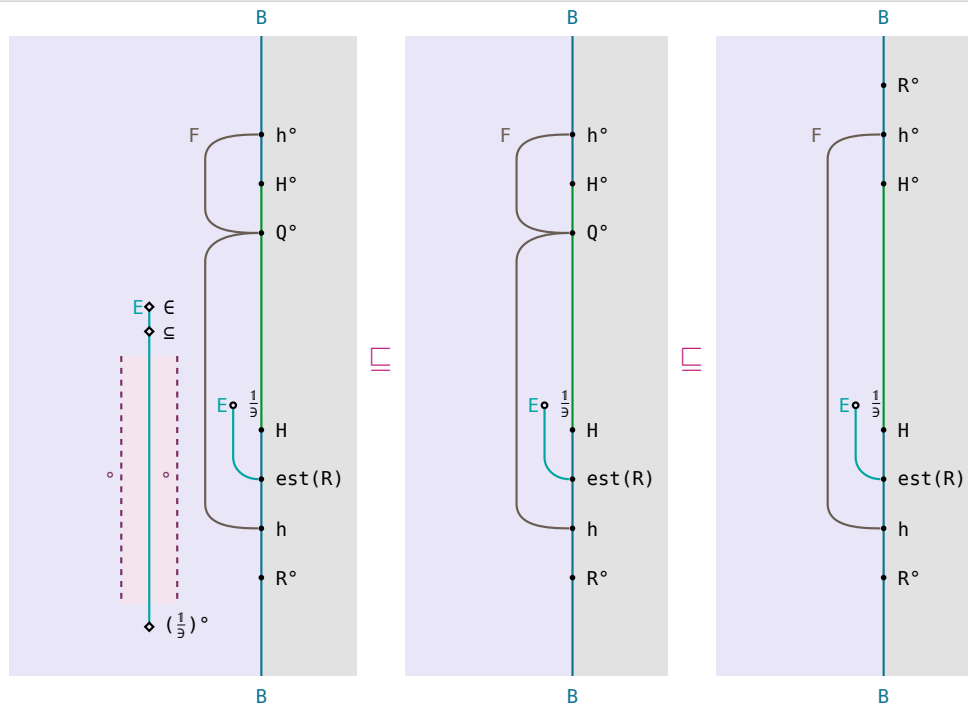
$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonotonicAlg}(h, R^\circ) \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge T^\circ F(H)h = H \wedge QF(H)h \sqsubseteq F(H)hR \implies H^\circ \frac{T^\circ}{\exists} \quad (15.2.1e)$$

$$\text{thin}(Q)P(F(\frac{H}{\exists} \text{est}(R))h)\text{est}(R) \sqsubseteq R^\circ$$

H° followed by the body at M is $\sqsubseteq R^\circ$: an answer of the body is never worse than an answer of H to the same input

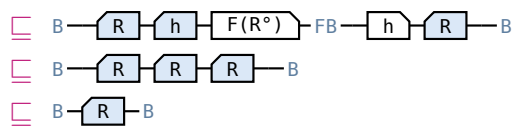
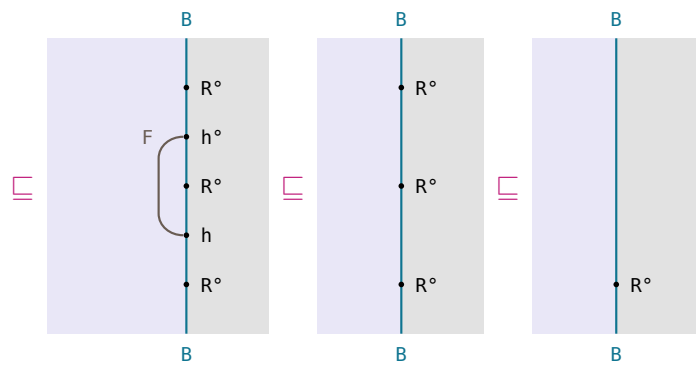


$$\begin{aligned} & B \xrightarrow{H} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} \text{thin}(Q) \xrightarrow{EFA} P(F(\frac{H}{\exists} \text{est}(R))h) \xrightarrow{EB} \text{est}(R) \xrightarrow{} B \\ \sqsubseteq & B \xrightarrow{H} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} \text{thin}(Q) \xrightarrow{EFA} \epsilon \backslash F(\frac{H}{\exists} \text{est}(R))hR^\circ \xrightarrow{} B \quad (9.4) \ P(X)\text{est}(R) \sqsubseteq \epsilon \backslash (XR^\circ) \text{ at } X \triangleq F(\frac{H}{\exists} \text{est}(R))h \text{ — est-710} \\ = & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{T} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} \text{thin}(Q) \xrightarrow{EFA} \epsilon \backslash F(\frac{H}{\exists} \text{est}(R))hR^\circ \xrightarrow{} B \\ & H^\circ = h^\circ F(H^\circ)T, \text{ the converse of } T^\circ F(H)h = H \text{ — hylo-fix} \\ \sqsubseteq & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{\epsilon} \text{thin}(Q) \xrightarrow{EFA} \epsilon \backslash F(\frac{H}{\exists} \text{est}(R))hR^\circ \xrightarrow{} B \quad T \frac{T^\circ}{\exists} \sqsubseteq \epsilon, \text{ not a tabulated row: } \frac{T^\circ}{\exists} \ni T^\circ \text{ conversed} \\ \sqsubseteq & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} Q \xrightarrow{FA} \epsilon \xrightarrow{EFA} \epsilon \backslash F(\frac{H}{\exists} \text{est}(R))hR^\circ \xrightarrow{} B \quad \epsilon \text{thin}(Q) \sqsubseteq Q^\circ \epsilon: \text{ a dropped candidate is } Q\text{-below a kept one — thin-laws} \end{aligned}$$



$\epsilon(\epsilon \setminus Y) \subseteq Y$, not a tabulated row: division cancels

$QF(H)h \subseteq F(H)hR$ conversed — the hypothesis on Q



$H^\circ M \subseteq R^\circ$ under F — est-up

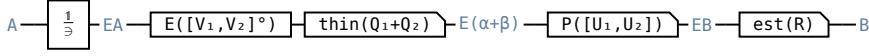
$h^\circ F(R^\circ)h \subseteq R^\circ$: h monotonic on R , shunted — the hypothesis on h

R transitive, twice

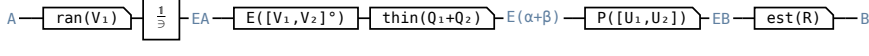
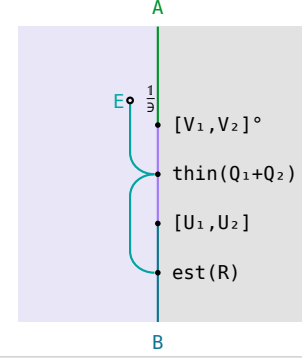
$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1 + Q_2) P([U_1, U_2]) \text{est}(R) = \text{ran}(V_1) \frac{V_1^\circ}{\exists} \text{thin}(Q_1) P(U_1) \text{est}(R) \cup \text{ran}(V_2) \frac{V_2^\circ}{\exists} \text{thin}(Q_2) P(U_2) \text{est}(R) \quad (15.2.1g)$$

when V_1 and V_2 have disjoint ranges, the body over $[V_1, V_2]$ runs the V_1 problem on inputs V_1 reaches and the V_2 problem on inputs V_2 reaches

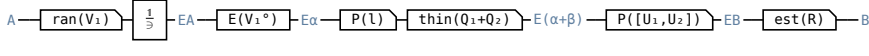
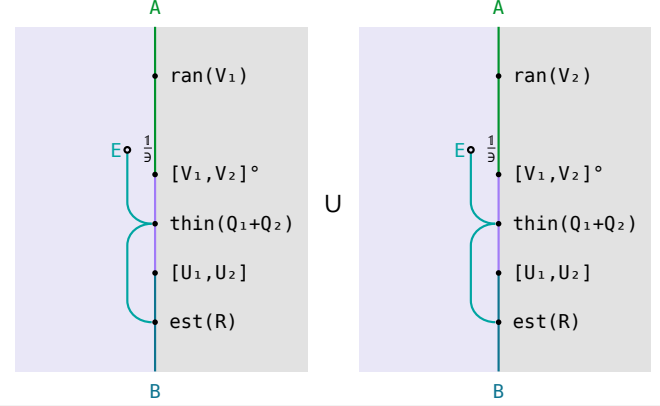
circuit



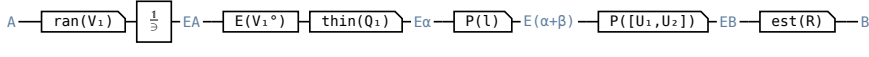
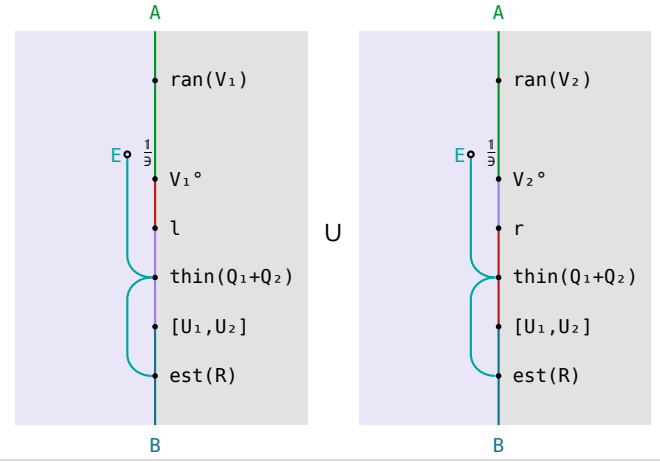
Hinze–Marsden



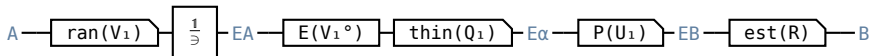
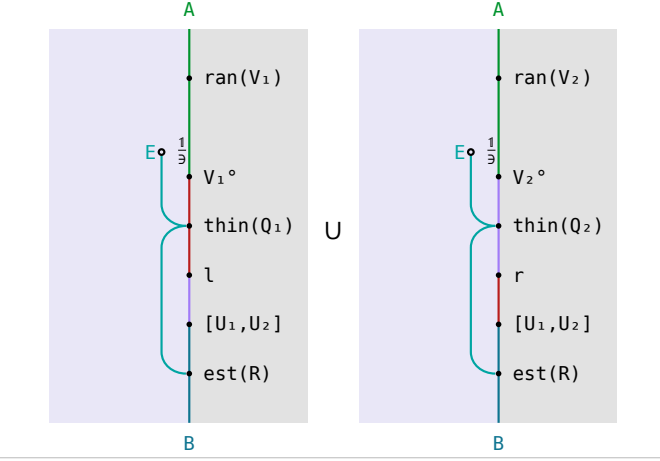
off $\text{ran } V_1 \cup \text{ran } V_2$ the set is empty and $\text{est}(R)$ of it is nothing; not a tabulated row



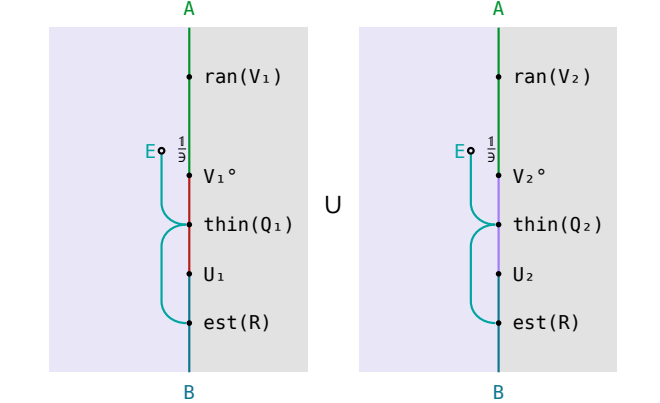
Exercise 9.5: on $\text{ran } V_1$, $\frac{[V_1, V_2]^\circ}{\exists} = \frac{V_1^\circ}{\exists} P(\text{inl})$, as V_1, V_2 have disjoint ranges



Exercise 9.5: $P(\text{inl}) \text{thin}(Q_1 + Q_2) = \text{thin}(Q_1) P(\text{inl})$

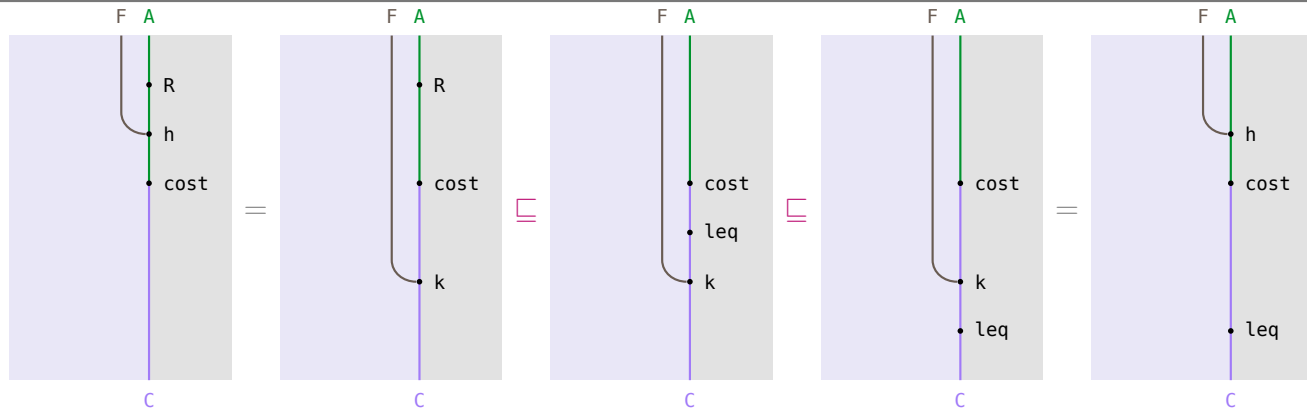


P a relator and $\text{inl}[U_1, U_2] = U_1$



$$\text{Map}(\text{cost}) \wedge R = \text{cost} \text{ leq } \text{cost}^\circ \wedge h \text{ cost} = F(\text{cost}) k \wedge F(\text{leq}) k \sqsubseteq k \text{ leq} \implies \text{MonotonicAlg}(h, R) \quad (15.2.1h)$$

if R compares two values by comparing their **costs** under $\leq$, and h then **cost** equals $F(\text{cost})$ then a k monotonic on $\leq$, then h is monotonic on R



$$\begin{aligned} & FA \xrightarrow{\quad} \boxed{F(R)} \xrightarrow{\quad} h \xrightarrow{\quad} A \xrightarrow{\quad} \boxed{\text{cost}} \xrightarrow{\quad} C \\ = & FA \xrightarrow{\quad} \boxed{F(R)} \xrightarrow{\quad} \boxed{F(\text{cost})} \xrightarrow{\quad} FC \xrightarrow{\quad} \boxed{k} \xrightarrow{\quad} C \\ \sqsubseteq & FA \xrightarrow{\quad} \boxed{F(\text{cost})} \xrightarrow{\quad} \boxed{F(\text{leq})} \xrightarrow{\quad} FC \xrightarrow{\quad} \boxed{k} \xrightarrow{\quad} C \\ \sqsubseteq & FA \xrightarrow{\quad} \boxed{F(\text{cost})} \xrightarrow{\quad} FC \xrightarrow{\quad} \boxed{k} \xrightarrow{\quad} \boxed{\text{leq}} \xrightarrow{\quad} C \\ = & FA \xrightarrow{\quad} \boxed{h} \xrightarrow{\quad} A \xrightarrow{\quad} \boxed{\text{cost}} \xrightarrow{\quad} \boxed{\text{leq}} \xrightarrow{\quad} C \end{aligned}$$

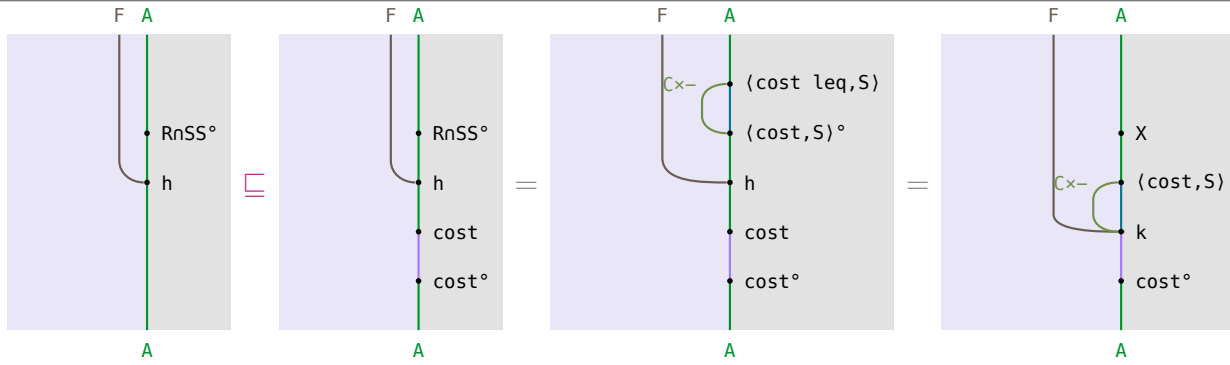
$F(R)h \sqsubseteq hR$ iff $F(R)h \text{ cost} \sqsubseteq h \text{ cost} \leq$: definition of R and shunting
 assumption $h \text{ cost} = F(\text{cost})k$
 $R \text{ cost} \sqsubseteq \text{cost} \leq$, as cost is a map; functors
 assumption $F(\leq)k \sqsubseteq k \leq$: k monotonic on $\leq$
 assumption $h \text{ cost} = F(\text{cost})k$

(15.2.1i)

the condition	how it is discharged
$F(R)h \sqsubseteq hR$ Proposition 9.2, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $h \text{ cost} = F(\text{cost})k$, $F(\leq)k \sqsubseteq k \leq$	monotonicity when the cost is itself a fold with a step k monotonic on $\leq$
$F(Rn(H^\circ H))h \sqsubseteq hR$ Proposition 9.3, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $h \text{ cost} = F(\langle \text{cost}, H^\circ \rangle)k$, $F(\leq \times 1)k \sqsubseteq k \leq$, H° simple;	monotonicity in context : k may also read the input the part was built from
$QF(H)h \sqsubseteq F(H)hR$ at $Q \triangleq F(U, V)$ Proposition 9.4, U, V preorders, $F(U, R)h \sqsubseteq hR$, $VH \sqsubseteq HR$	both conditions at once, split along the two arguments of a bifunctor

$$\text{Map}(\text{cost}) \wedge \text{Simple}(S) \wedge R = \text{cost} \text{ leq } \text{cost}^\circ \wedge h \text{ cost} = F(\langle \text{cost}, S \rangle) k \wedge F(\text{leq} \times 1) k \sqsubseteq k \text{ leq} \Rightarrow F(RnSS^\circ) h \sqsubseteq h R \quad (15.2.1j)$$

improving each part by R within its S -context, then assembling by h , is below hR

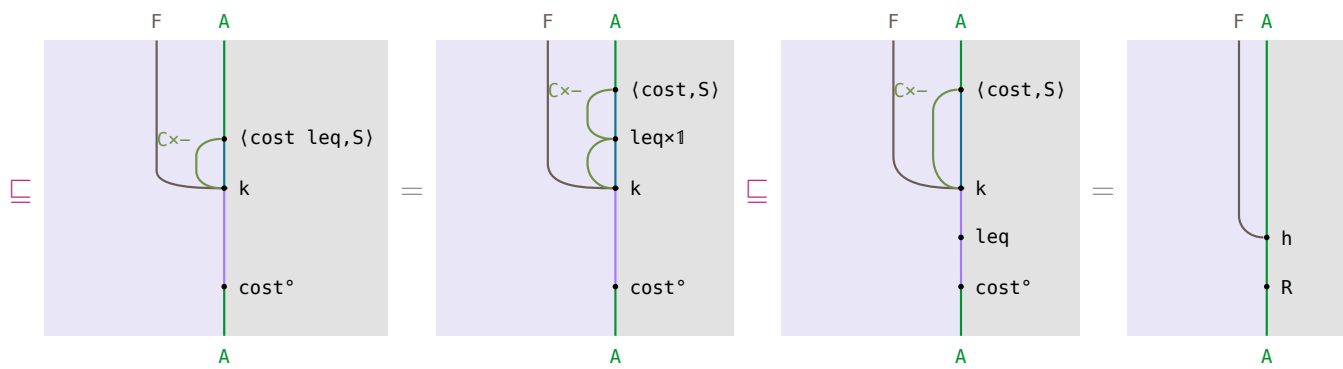


$$\begin{aligned} & FA \rightarrow \boxed{F(RnSS^\circ)} \rightarrow h \rightarrow A \\ \sqsubseteq & FA \rightarrow \boxed{F(RnSS^\circ)} \rightarrow h \rightarrow A \rightarrow \boxed{\text{cost}} \rightarrow C \rightarrow \boxed{\text{cost}} \rightarrow A \\ = & FA \rightarrow \boxed{F(\langle \text{cost leq}, S \rangle \langle \text{cost}, S \rangle^\circ)} \rightarrow h \rightarrow A \rightarrow \boxed{\text{cost}} \rightarrow C \rightarrow \boxed{\text{cost}} \rightarrow A \\ = & FA \rightarrow \boxed{F(X)} \rightarrow \boxed{F(\langle \text{cost}, S \rangle)} \rightarrow F(C \times B) \rightarrow k \rightarrow C \rightarrow \boxed{\text{cost}} \rightarrow A \end{aligned}$$

shunting: cost a map, so $1 \sqsubseteq \text{cost} \text{ cost}^\circ$ — triple-chains

products: $RnSS^\circ = \langle \text{cost leq}, S \rangle \langle \text{cost}, S \rangle^\circ$ at $R = \text{cost leq cost}^\circ$ — $\triangle \times$

assumption on cost : $h \text{ cost} = F(\langle \text{cost}, S \rangle) k$



$$\begin{aligned} \sqsubseteq & FA \rightarrow \boxed{F(\langle \text{cost leq}, S \rangle)} \rightarrow F(C \times B) \rightarrow k \rightarrow C \rightarrow \boxed{\text{cost}} \rightarrow A \\ = & FA \rightarrow \boxed{F(\langle \text{cost}, S \rangle)} \rightarrow \boxed{F(\text{leq} \times 1)} \rightarrow F(C \times B) \rightarrow k \rightarrow C \rightarrow \boxed{\text{cost}} \rightarrow A \\ \sqsubseteq & FA \rightarrow \boxed{F(\langle \text{cost}, S \rangle)} \rightarrow F(C \times B) \rightarrow k \rightarrow \boxed{\text{leq}} \rightarrow C \rightarrow \boxed{\text{cost}} \rightarrow A \\ = & FA \rightarrow h \rightarrow R \rightarrow A \end{aligned}$$

S simple, so $\langle \text{cost}, S \rangle$ simple: $\langle \text{cost}, S \rangle^\circ \langle \text{cost}, S \rangle \sqsubseteq 1$

products; functors: $\langle \text{cost leq}, S \rangle = \langle \text{cost}, S \rangle (\text{leq} \times 1)$ — bdm-prod-laws, relator-laws

assumption on k : $F(\text{leq} \times 1) k \sqsubseteq k \text{ leq}$

assumption on cost read backwards, then $R = \text{cost leq cost}^\circ$

$\mathbf{G(U,R)h \sqsubseteq hR \wedge Vh \sqsubseteq HR \implies G(U,V)G(1,H)h \sqsubseteq G(1,H)hR}$ (15.2.1k)	
thinning the parts by $\mathbf{U}$ in the first argument and by $\mathbf{V}$ in the second $(\mathbf{G(U,V)})$, then solving by $\mathbf{H}$ and assembling by $\mathbf{h}$, gives only what solving and assembling and then improving by $\mathbf{R}$ gives $(\mathbf{G(1,H)hR})$	
circuit	Hinze–Marsden
$G\ e\ w \text{---} \boxed{G(U,V)} \text{---} \boxed{G(1,H)} \text{---} G\ e\ A \text{---} \boxed{h} \text{---} A$	
$= G\ e\ w \text{---} \boxed{G(U,VH)} \text{---} G\ e\ A \text{---} \boxed{h} \text{---} A$ taking $Q \triangleq G(U,V)$; bifunctors: $G(U,V)G(1,H) = G(U,VH)$ — relator-laws	
$\sqsubseteq G\ e\ w \text{---} \boxed{G(U,HR)} \text{---} G\ e\ A \text{---} \boxed{h} \text{---} A$ assumption on V: $Vh \sqsubseteq HR$	
$= G\ e\ w \text{---} \boxed{G(1,H)} \text{---} \boxed{G(U,R)} \text{---} G\ e\ A \text{---} \boxed{h} \text{---} A$ bifunctors: $G(U,HR) = G(1,H)G(U,R)$ — relator-laws	
$\sqsubseteq G\ e\ w \text{---} \boxed{G(1,H)} \text{---} G\ e\ A \text{---} \boxed{h} \text{---} \boxed{R} \text{---} A$ assumption on h: $G(U,R)h \sqsubseteq hR$	

§15.3. The string edit problem

(15.3a)

$Op ::= \text{cpy Char} \mid \text{del Char} \mid \text{ins Char}, F(A,B) = 1 + (A \times B), \alpha \triangleq [\text{nil}, \text{cons}].$

$\text{edit} \triangleq ([\text{base}, \text{step}]) : [Op] \rightarrow [\text{Char}] \times [\text{Char}],$ base returning $([], [])$, step $(\text{cpy } a, (xs, ys)) = ([a] \# xs, [a] \# ys)$, step $(\text{del } a, (xs, ys)) = ([a] \# xs, ys)$, step $(\text{ins } a, (xs, ys)) = (xs, [a] \# ys)$.

$\text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}]), R \triangleq \text{length} \leq \text{length}^\circ, U \triangleq \top, V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ, Q \triangleq 1 + (U \times V),$ empty the coreflexive on (xs, ys) with both lists empty.

unstep implements $\frac{\text{step}^\circ}{\exists} \text{thin}(U \times V) : \text{unstep } ([a] \# xs, []) = [(\text{del } a, (xs, []))], \text{unstep } ([], [b] \# ys) = [(\text{ins } b, ([], ys))], \text{unstep } ([a] \# xs, [b] \# ys) = (a = b \rightarrow [\text{cpy } a, (xs, ys)], [(\text{del } a, (xs, [b] \# ys)), (\text{ins } b, ([a] \# xs, ys))]).$

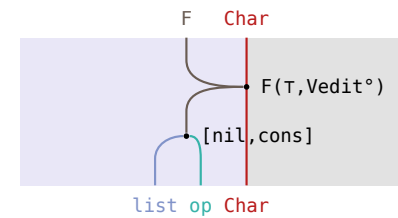
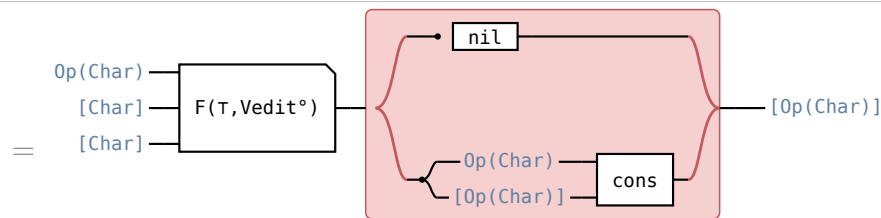
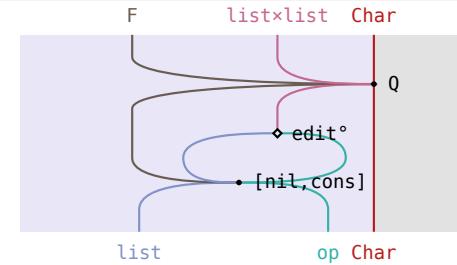
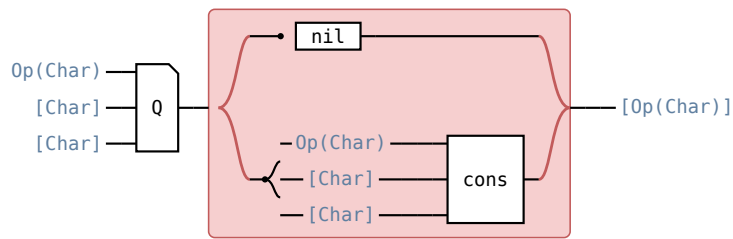
$$QF(\text{edit}^\circ)[\text{nil}, \text{cons}] \sqsubseteq F(\text{edit}^\circ)[\text{nil}, \text{cons}]R$$

(15.3d)

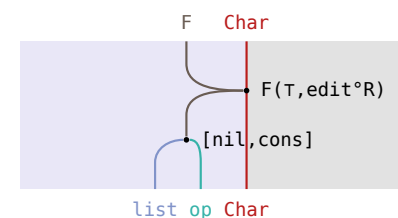
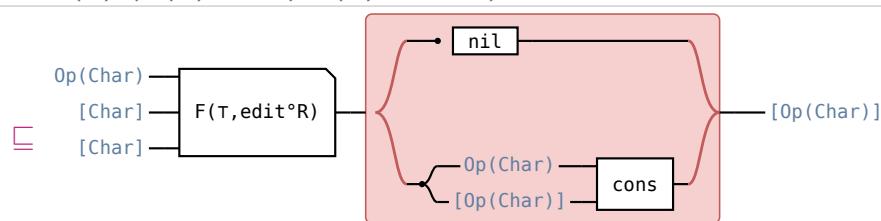
every edit sequence built from a decomposition Q puts above a given one — any operation, each string lengthened at the front — is at least as long as one built from the given one

circuit

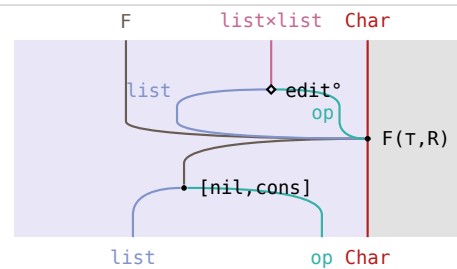
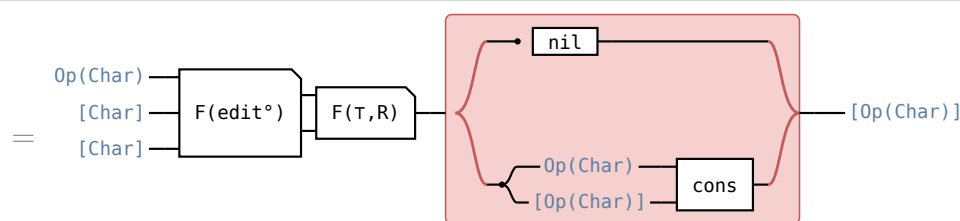
Hinze–Marsden



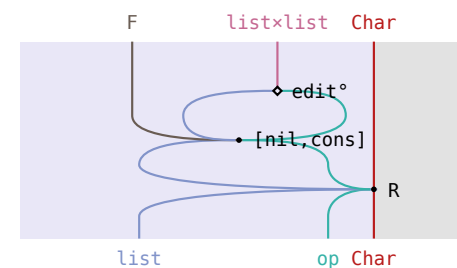
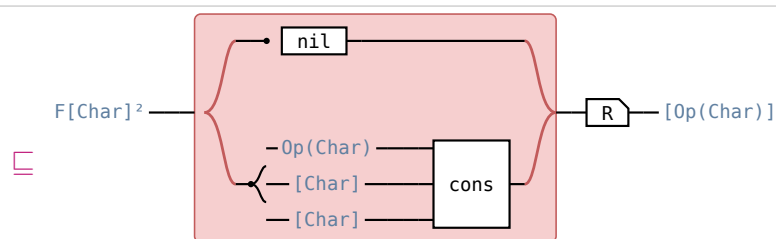
$Q \triangleq 1 + (U \times V)$ is $F(U, V)$ at $U \triangleq \tau$ — $\triangleq \text{edit}$; and the bifunctor F preserves composition: $F(U, V)F(1, \text{edit}^\circ) = F(U, V \text{ edit}^\circ)$



Proposition 9.4's second condition $V \text{ edit}^\circ \sqsubseteq \text{edit}^\circ R$ — (15.3e)



F preserves composition: $F(U, \text{edit}^\circ R) = F(1, \text{edit}^\circ)F(U, R)$



Proposition 9.4's first condition $F(\tau, R) \alpha \sqsubseteq \alpha R$, left as an exercise in the book: **CONS** adds one to both lengths whatever the two operations are

$\text{edit } V^\circ \sqsubseteq R^\circ \text{ edit}$ (15.3e) cutting a front off either string an edit sequence produces ($\text{edit } V^\circ$) leaves a pair that a sequence no longer produces ($R^\circ \text{ edit}$)	
circuit	Hinze–Marsden
$=$ $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$, and $\times$ preserves composition	
$\sqsubseteq$ $\text{edit } (\text{suffix} \times 1) \sqsubseteq R^\circ \text{ edit}$: drop the operation that produced the head, or weaken its cpy to an ins. Proved by induction at suffix directly, not through $\text{suffix} = \text{tail}^*$ and $BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B$ as the book does	
$\sqsubseteq$ $\text{edit } (1 \times \text{suffix}) \sqsubseteq R^\circ \text{ edit}$, the mirror image: cpy weakens to a del	
$\sqsubseteq$ R° is transitive, $\leq$ being so	

$\text{step } \text{base}^\circ = \emptyset$ (15.3f)	
no pair of strings is both a result of step and the result $([], [])$ of base	
circuit	Hinze–Marsden
base returns only $([], [])$, so $\text{base}^\circ = \text{empty } \text{base}^\circ$	
$\text{step } \text{empty} = \emptyset$: cpy and del put a character on the left string, ins one on the right	
$\emptyset$ composed with anything is $\emptyset$	

(15.3g)

$\text{mle}(xs, ys) = \text{head}(\text{column}(xs, ys))$, $\text{column}(xs, ys) = [\text{mle}(u, ys) \mid u \leftarrow \text{tails}(xs)]$ $= \text{column}(xs) = ([\text{fstcol}(xs), \text{nextcol}(xs)])$, $\text{fstcol} = \text{list}(\text{del}) \text{ tails}$ the tabulation: $\text{mle}(xs, ys)$ needs $\text{mle}(u, v)$ for every tail u of xs and v of ys , so the columns are built right to left	
$\text{column}(xs, \text{cons}(b, ys)) = \text{nextcol}(xs, (b, \text{column}(xs, ys)))$ $= \text{nextcol}(xs)(b, us) = ([\text{base}(b, \text{last}(us)), \text{step}(b)])(xus)$, $xus = \text{zip}(xs, \text{zip}(\text{init}(us), \text{tail}(us)))$ each column is a fold built bottom to top, over xs zipped with the adjacent pairs of the column to its right	
$\text{base}(b, u) = [[\text{ins}(b)] \# u]$ $= \text{step}(b)((a, (u, v)), ws) = (a \rightarrow b \rightarrow [[\text{cpy}(a)] \# v] \# ws, [\text{bmin}(R)([\text{del}(a)] \# w, [\text{ins}(b)] \# u)] \# ws)$ $w = \text{head}(ws)$; these base , step are not edit 's. An entry depends on the one below it (a delete), the one to its right (an insert), and the one below that (a copy) — quadratic in the two lengths	

§15.4. Optimal bracketing

(15.4a)

$\text{tree } A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)$, $FX = A + X^2$, so $F(R) = 1 + R^2$; $h \triangleq [\text{tip}, \text{bin}]$, $\text{flatten} \triangleq ([\text{wrap}, \text{cat}]) : \text{tree } A \rightarrow \text{list}^+ A$, $H = \text{flatten}^\circ$. $\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}])$, $\text{opt} \triangleq (\text{zero}, \text{st})$, $\text{opb } ((cx, sx), (cy, sy)) = (\text{cb } (sx, sy) + cx + cy, \text{sb } (sx, sy))$. sb associative, so $\text{size} = \text{flatten } sz$ for a map sz $R \triangleq \text{cost} \leq \text{cost}^\circ$, $g \triangleq [\text{zero}, (1 \times sz)^2 \text{ opb } \pi_1]$, single the coreflexive on singleton lists. $\text{splits} \triangleq (\text{inits}^+, \text{tails}^+) \text{ zip}$, an implementation of $\frac{\text{cat}^\circ}{\Sigma}$; $\text{array} \triangleq \text{inits } \text{list}(\text{row})$, $\text{row} \triangleq \text{tails } \text{list}(\text{mct})$, $\text{col} \triangleq \text{inits } \text{list}(\text{mct})$. $\text{mix} \triangleq \text{zip } \text{list}(\text{bin}) \text{ minlist}(R)$, $\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc}$, $\text{process} \triangleq ((\text{tip } \text{wrap}) \times 1) \text{ loop}(\text{next})$.	
--	--

$\text{Assoc}(\text{sb}) \Rightarrow F(\langle \text{cost}, \text{flatten} \rangle)g = [\text{tip}, \text{bin}] \text{cost}$ (15.4b)	
pairing each subtree with its cost and its flattening and then applying g gives what building the tree by $[\text{tip}, \text{bin}]$ and taking its cost gives	
circuit	Hinze–Marsden
$F(\text{tree}(A)) \rightarrow F(\langle \text{cost}, \text{flatten} \rangle) \rightarrow F(\text{Int} \times \text{list}^+(A)) \rightarrow g \rightarrow \text{Int}$	
$A + \text{tree}(A)^2 \rightarrow \text{zero} \rightarrow \text{cost} \rightarrow \text{flatten} \rightarrow \text{list}^+(A) \rightarrow \text{sz} \rightarrow \text{opb} \rightarrow \text{Int}$ definition of g; coproducts and products — $\triangleq \text{flatten}$	
$A + \text{tree}(A)^2 \rightarrow \text{zero} \rightarrow \text{cost} \rightarrow \text{size} \rightarrow \text{opb} \rightarrow \text{Int}$ $\text{flatten sz} = \text{size}$, since sb is associative — $\triangleq \text{flatten}$	
$A + \text{tree}(A)^2 \rightarrow \text{zero} \rightarrow \text{bin} \rightarrow \text{cost} \rightarrow \text{size} \rightarrow \text{opb} \rightarrow \text{Int}$ $\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}])$, at a node $\text{bin}(\text{cost}, \text{size}) = (\text{cost}, \text{size})^2 \text{ opb}$	
$A + \text{tree}(A)^2 \rightarrow \text{tip} \rightarrow \text{tree}(A) \rightarrow \text{cost} \rightarrow \text{bin} \rightarrow \text{tree}(A) \rightarrow \text{cost} \rightarrow \text{Int}$ $\text{tip cost} = \text{zero}$; products, $\langle \text{cost}, \text{size} \rangle \pi_1 = \text{cost}$	
$A + \text{tree}(A)^2 \rightarrow \text{tip} \rightarrow \text{tree}(A) \rightarrow \text{cost} \rightarrow \text{bin} \rightarrow \text{tree}(A) \rightarrow \text{cost} \rightarrow \text{Int}$ coproducts, $[\text{tip cost}, \text{bin cost}] = [\text{tip}, \text{bin}] \text{cost}$	

$F(\leq \times 1)g \sqsubseteq g \leq$ (15.4c)	
raising the cost of either subtree ($\leq \times 1$ in both slots) and then applying g returns a value at least what g returns on the costs before raising ($g \leq$)	
circuit	Hinze–Marsden
$F(\text{Int} \times \text{list}^+(A)) \rightarrow \boxed{F(\leq \times 1)} \rightarrow \boxed{g} \rightarrow \text{Int}$	
$A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$ $=$	
$A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$ $\sqsubseteq$	
$A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$ $\sqsubseteq$	
$\leq$ reflexive, so $\text{zero} \sqsubseteq \text{zero} \leq$; coproducts	
$F(\text{Int} \times \text{list}^+(A)) \rightarrow \boxed{g} \rightarrow \boxed{\leq} \rightarrow \text{Int}$ $=$ definition of $g \triangleq \text{flatten}$	

$$\text{Assoc}(\text{sb}) \Rightarrow (\mu X : \frac{[\text{wrap}, \text{cat}]^\circ}{3} P([\text{tip}, (X \times X) \text{bin}]) \text{est}(R)) \sqsubseteq \frac{\text{flatten}^\circ}{3} \text{est}(R), \text{ mct} = (\text{single} \rightarrow \text{head tip}, \text{ (init col, tail row) mix}) \quad (15.4d)$$

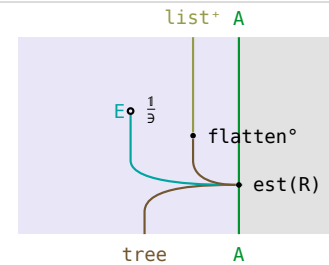
a least-cost bracketing of $a_1 \oplus \dots \oplus a_n$ is read off an array holding one best tree per non-empty segment, each entry built from the column to its left and the row below it

circuit

$$\text{list}^+(A) \xrightarrow{\frac{1}{3}} E(\text{list}^+(A)) \xrightarrow{E(\text{flatten}^\circ)} E(\text{tree}(A)) \xrightarrow{\text{est}(R)} \text{tree}(A)$$

the specification $\text{---} \triangleq \text{flatten}$

Hinze–Marsden



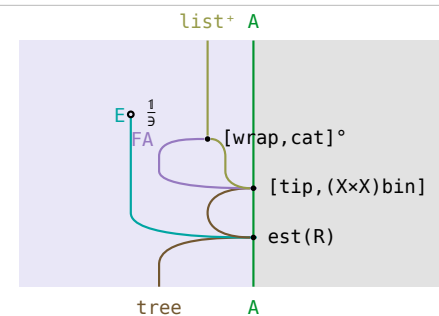
$$\text{list}^+(A) \xrightarrow{\frac{1}{3}} E(\text{list}^+(A)) \xrightarrow{E([\text{wrap}, \text{cat}]^\circ)} E(F(\text{list}^+(A))) \xrightarrow{P([\text{tip}, (X \times X) \text{bin}])} E(\text{tree}(A)) \xrightarrow{\text{est}(R)} \text{tree}(A)$$

split the list in every way, bracket both halves, join. The condition is monotonicity **in context**,

$\sqsubseteq F(Rn(\text{flatten} \text{ flatten}^\circ)) h \sqsubseteq hR$ — only trees with the same flattening are compared —

which is Proposition 9.3 at $H^\circ = \text{flatten}$, a map, with (9.5) $[\text{tip}, \text{bin}]$

$\text{cost} = (1 + (\text{cost}, \text{flatten})^2)g$ (the cost of a node reads only the cost and the flattening of its two subtrees) and (9.6) $(1 + (\leq 1)^2)g \sqsubseteq g$ (g monotonic on $\leq$ in its two cost arguments)

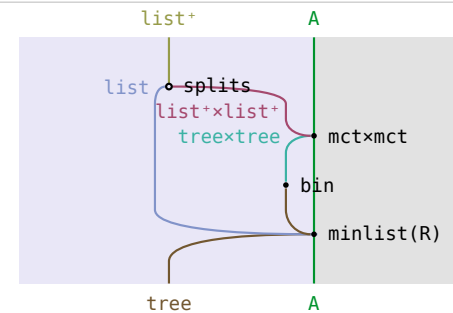


$$\text{list}^+(A) \xrightarrow{\frac{1}{3}} E(\text{list}^+(A)) \xrightarrow{E(\text{cat}^\circ)} E(\text{list}^+(A)^2) \xrightarrow{P((X \times X) \text{bin})} E(\text{tree}(A)) \xrightarrow{\text{est}(R)} \text{tree}(A)$$

$\equiv$ Proposition 9.1: **wrap** and **cat** have disjoint ranges, and **single** is the coreflexive on singleton lists, where **wrap** returns

$$\text{list}^+(A) \xrightarrow{\text{splits}} [\text{list}^+(A)^2] \xrightarrow{\text{list}((\text{mct} \times \text{mct}) \text{bin})} [\text{tree}(A)] \xrightarrow{\text{minlist}(R)} \text{tree}(A)$$

$\sqsubseteq \text{splits} \triangleq \langle \text{inits}^+, \text{tails}^+ \rangle$ zip implements $\frac{\text{cat}^\circ}{3}$ and $\text{minlist}(R)$ implements $\text{est}(R)$. Exponential, since the segments of one list overlap



$$\text{mct} = (\text{single} \rightarrow \text{head tip}, (\text{init col}, \text{tail row}) \text{mix}) \quad (9.7)$$

$\equiv$ the tabulation: **mct** $\times S$ is needed for every non-empty segment $\times S$, so the values are held as an array of rows, $\text{array} \triangleq \text{inits list}(\text{row}), \text{row} \triangleq \text{tails list}(\text{mct}), \text{col} \triangleq \text{inits list}(\text{mct}), \text{mix} \triangleq \text{zip list}(\text{bin}) \text{minlist}(R)$

$$\text{col} = (\text{single} \rightarrow \text{head tip wrap}, (\text{init col}, \text{tail row}) \text{next}) \quad (9.8)$$

$$\text{cons col} = (1 \times \text{array}) \text{process} \quad (9.9)$$

$$\equiv \text{row} = (\text{single} \rightarrow \text{head tip wrap}, (\text{mct}, \text{tail row}) \text{cons}) \quad (9.10)$$

a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop, $\text{next} \triangleq (\pi_1, \text{mix}) \text{snoc}, \text{process} \triangleq ((\text{tip wrap}) \times 1) \text{loop}(\text{next})$

$$\text{array} = ([\text{fstcol}, \text{addcol}]), \text{fstcol} \triangleq \text{tip wrap wrap}$$

$$\equiv \text{addcol} \triangleq (\pi_1 \text{tip wrap}, \text{step}) \text{cons}, \text{step} \triangleq (\text{process tail}, \pi_2) \text{zip list}(\text{cons})$$

the program: one fold building the array column by column, cubic in the length of the input

nonsingle mct=nonsingle(init col,tail row)mix

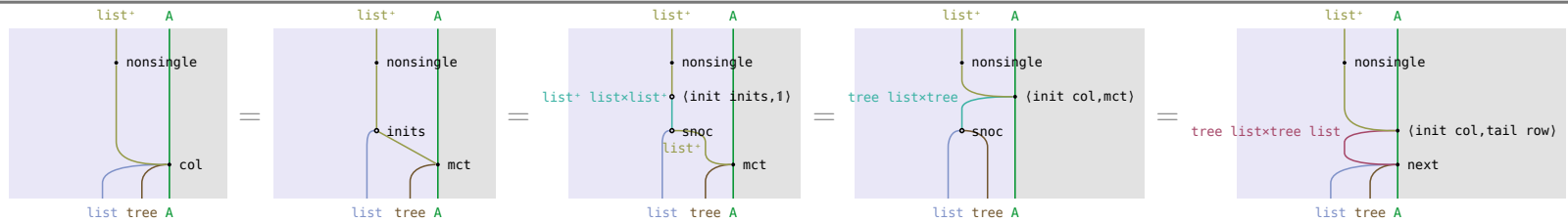
(15.4e)

on a list of two or more elements, **mct** gives what **mix** gives on the column of its **init** and the row of its **tail**

$$\begin{aligned}
 & \text{list}^+(A) \text{---} \boxed{\text{nonsingle}} \text{---} \boxed{\text{mct}} \text{---} \text{tree}(A) \\
 = & \text{list}^+(A) \text{---} \boxed{\text{nonsingle}} \text{---} \boxed{\text{splits}} \text{---} [\text{list}^+(A)^2] \text{---} \boxed{\text{list}(\text{graft})} \text{---} [\text{tree}(A)] \text{---} \boxed{\text{minlist}(R)} \text{---} \text{tree}(A) && \text{recursive case of mct} \text{---} \triangleq \text{flatten} \\
 = & \text{list}^+(A) \text{---} \boxed{\text{nonsingle}} \text{---} \begin{cases} \boxed{\text{inits}^+} \\ \boxed{\text{tails}^+} \end{cases} \text{---} [\text{list}^+(A)] \text{---} \boxed{\text{zip}} \text{---} [\text{list}^+(A)^2] \text{---} \boxed{\text{list}(\text{graft})} \text{---} [\text{tree}(A)] \text{---} \boxed{\text{minlist}(R)} \text{---} \text{tree}(A) \\
 & \text{definition of splits} \text{---} \triangleq \text{flatten} \\
 = & \text{list}^+(A) \text{---} \boxed{\text{nonsingle}} \text{---} \boxed{\text{trees}} \text{---} \boxed{\text{zip}} \text{---} [\text{tree}(A)^2] \text{---} \boxed{\text{list}(\text{bin})} \text{---} [\text{tree}(A)] \text{---} \boxed{\text{minlist}(R)} \text{---} \text{tree}(A) \\
 & \text{definitions of } \text{graft} \triangleq (\text{mct} \times \text{mct}) \text{ bin and } \text{trees} \triangleq (\text{inits}^+ \text{ list}(\text{mct}), \text{tails}^+ \text{ list}(\text{mct})), \text{ our names; zip} \\
 & \text{list}(f \times g) = (\text{list}(f) \times \text{list}(g)) \text{ zip} \\
 = & \text{list}^+(A) \text{---} \boxed{\text{nonsingle}} \text{---} \boxed{\text{trees}} \text{---} \boxed{\text{mix}} \text{---} \text{tree}(A) && \text{introducing } \text{mix} \triangleq \text{zip list}(\text{bin}) \text{ minlist}(R) \text{---} \triangleq \text{flatten} \\
 = & \text{list}^+(A) \text{---} \boxed{\text{nonsingle}} \text{---} \begin{cases} \boxed{\text{init}} \text{---} \boxed{\text{col}} \\ \boxed{\text{tail}} \text{---} \boxed{\text{row}} \end{cases} \text{---} [\text{tree}(A)] \text{---} \boxed{\text{mix}} \text{---} \text{tree}(A) \\
 & \text{definition of trees; } \text{inits}^+ = \text{init inits and tails}^+ = \text{tail tails on non-singletons; definition of row and col} \text{---} \triangleq \text{flatten}
 \end{aligned}$$

nonsingle col=nonsingle(init col,tail row)next

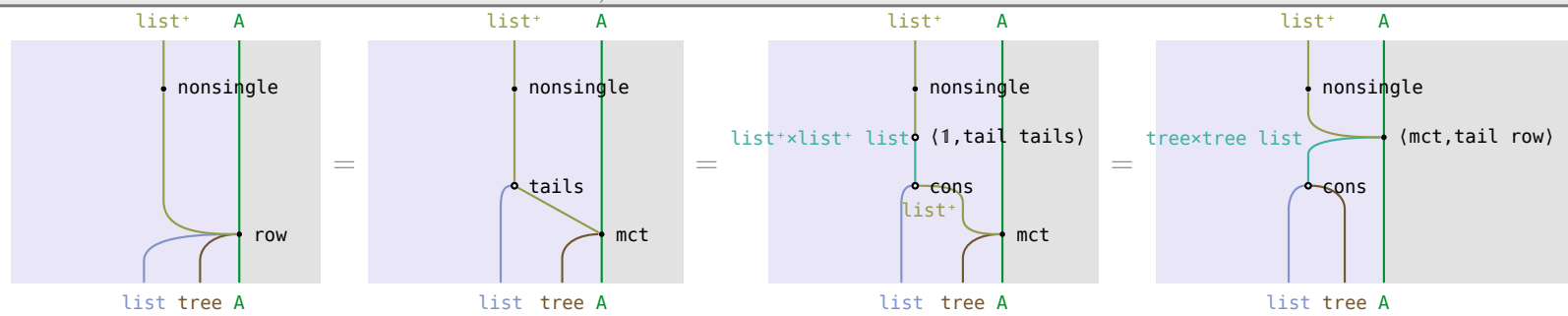
(15.4f)

on a list of two or more elements, the column is what **next** makes of the column of its **init** and the row of its **tail**

$$\begin{aligned}
 & \text{list}^+(A) \text{---} \boxed{\text{nonsingle}} \text{---} \boxed{\text{col}} \text{---} [\text{tree}(A)] \\
 = & \text{list}^+(A) \text{---} \boxed{\text{nonsingle}} \text{---} \boxed{\text{inits}} \text{---} [\text{list}^+(A)] \text{---} \boxed{\text{list}(\text{mct})} \text{---} [\text{tree}(A)] && \text{definition of col} \text{---} \triangleq \text{flatten} \\
 = & \text{list}^+(A) \text{---} \boxed{\text{nonsingle}} \text{---} \begin{cases} \boxed{\text{init}} \text{---} \boxed{\text{inits}} \\ \text{list}^+(A) \end{cases} \text{---} \boxed{\text{snoc}} \text{---} [\text{list}^+(A)] \text{---} \boxed{\text{list}(\text{mct})} \text{---} [\text{tree}(A)] \\
 & \text{inits} = (\text{init inits}, 1) \text{ snoc on non-singletons} \\
 = & \text{list}^+(A) \text{---} \boxed{\text{nonsingle}} \text{---} \begin{cases} \boxed{\text{init}} \text{---} \boxed{\text{col}} \\ \boxed{\text{mct}} \end{cases} \text{---} [\text{tree}(A)] \text{---} \boxed{\text{snoc}} \text{---} [\text{tree}(A)] && \text{snoc list}(f) = (\text{list}(f) \times f) \text{ snoc; definition of col} \\
 = & \text{list}^+(A) \text{---} \boxed{\text{nonsingle}} \text{---} \begin{cases} \boxed{\text{init}} \text{---} \boxed{\text{col}} \\ \boxed{\text{tail}} \text{---} \boxed{\text{row}} \end{cases} \text{---} [\text{tree}(A)] \text{---} \boxed{\text{next}} \text{---} [\text{tree}(A)] \\
 & (9.7) \text{ on non-singletons} \text{---} (15.4e) \text{---} \text{and introducing } \text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc} \text{---} \triangleq \text{flatten}
 \end{aligned}$$

nonsingle row=nonsingle(mct,tail row)cons

(15.4g)

on a list of two or more elements, the row is **mct** of the whole list consed onto the row of its **tail**

$$\begin{aligned}
 & \text{list}^+(A) \text{---} \text{nonsingle} \text{---} \text{row} \text{---} [\text{tree}(A)] \\
 = & \text{list}^+(A) \text{---} \text{nonsingle} \text{---} \text{tails} \text{---} [\text{list}^+(A)] \text{---} \text{list}(\text{mct}) \text{---} [\text{tree}(A)] && \text{definition of row} \triangleq \text{flatten} \\
 = & \text{list}^+(A) \text{---} \text{nonsingle} \text{---} \begin{cases} \text{tail} \text{---} \text{tails} \text{---} [\text{list}^+(A)] \\ \text{mct} \text{---} \text{tree}(A) \end{cases} \text{---} \text{cons} \text{---} [\text{list}^+(A)] \text{---} \text{list}(\text{mct}) \text{---} [\text{tree}(A)] \\
 & \text{tails} = (1, \text{tail tails}) \text{ cons on non-singletons} \\
 = & \text{list}^+(A) \text{---} \text{nonsingle} \text{---} \begin{cases} \text{mct} \text{---} \text{tree}(A) \\ \text{tail} \text{---} \text{row} \text{---} [\text{tree}(A)] \end{cases} \text{---} \text{cons} \text{---} [\text{tree}(A)] && \text{cons list}(f) = (f \times \text{list}(f)) \text{ cons; definition of row}
 \end{aligned}$$

cons col=(1×array)process

(15.4h)

the column of **a** consed onto **x** is what **process** makes of **a** and the array of **x****circuit****Hinze–Marsden**

$\text{list}^+(A) \text{---} \text{cons} \text{---} \text{list}^+(A) \text{---} \text{col} \text{---} [\text{tree}(A)]$	
$ \begin{aligned} & \text{list}^+(A) \text{---} \text{tip} \text{---} \text{tree}(A) \text{---} \text{wrap} \text{---} \text{loop}(\text{next}) \text{---} [\text{tree}(A)] \\ = & \text{Exercise 9.13 with } k \triangleq \text{cons col}, g \triangleq \text{tip wrap}, h \triangleq \text{row}, f \triangleq \text{next: its two equations are (9.8)} \\ & \text{--- (15.4f) --- at } a: [b] \text{ and at } a: (u++[b]), \text{ with } [a] \text{ col} = [\text{tip}(a)] \end{aligned} $	
$ \begin{aligned} & \text{list}^+(A) \text{---} \text{array} \text{---} \text{tip} \text{---} \text{tree}(A) \text{---} \text{wrap} \text{---} \text{loop}(\text{next}) \text{---} [\text{tree}(A)] \\ = & \text{products} \end{aligned} $	
$ \begin{aligned} & \text{list}^+(A) \text{---} \text{array} \text{---} \text{process} \text{---} [\text{tree}(A)] \\ = & \text{definition of process} \end{aligned} $	

tops=(1×array)process tail (15.4i)	
the trees at the top of the new rows are what process makes of a and the array of x , less its first	
circuit	Hinze–Marsden
$\text{list}^+(A) \xrightarrow{A} \boxed{\text{tops}} \rightarrow [\text{tree}(A)]$	
$= \text{list}^+(A) \xrightarrow{A} \boxed{\text{cons}} \rightarrow \text{list}^+(A) \xrightarrow{\text{init}} \boxed{\text{tail}} \rightarrow [\text{list}^+(A)] \xrightarrow{\text{list(mct)}} \boxed{\text{list(mct)}} \rightarrow [\text{tree}(A)]$ definition of $\text{tops} \triangleq \text{tic list(mct)}$, our name	
$= \text{list}^+(A) \xrightarrow{A} \boxed{\text{cons}} \rightarrow \text{list}^+(A) \xrightarrow{\text{col}} \boxed{\text{tail}} \rightarrow [\text{tree}(A)]$ $\text{tail list}(f) = \text{list}(f) \text{ tail}$; definition of col	
$= \text{list}^+(A) \xrightarrow{A} \boxed{\text{array}} \rightarrow \boxed{\text{process}} \rightarrow \boxed{\text{tail}} \rightarrow [\text{tree}(A)]$ (9.9) — (15.4h)	

rests=π_2 array (15.4j)	
the rest of each new row is the array of x	
circuit	Hinze–Marsden
$\text{list}^+(A) \xrightarrow{A} \boxed{\text{rests}} \rightarrow [[\text{tree}(A)]]$	
$= \text{list}^+(A) \xrightarrow{A} \boxed{\text{cons}} \rightarrow \text{list}^+(A) \xrightarrow{\text{init}} \boxed{\text{tail}} \rightarrow [\text{list}^+(A)] \xrightarrow{\text{list(tail row)}} \boxed{\text{list(tail row)}} \rightarrow [[\text{tree}(A)]]$ definition of $\text{rests} \triangleq \text{tic list(tail row)}$, our name	
$= \text{list}^+(A) \xrightarrow{A} \text{list}^+(A) \xrightarrow{\text{array}} \boxed{\text{array}} \rightarrow [[\text{tree}(A)]]$ $\text{tic list(tail)} = \pi_2 \text{ init}$; definition of array	

$\text{newrows} = (1 \times \text{array}) \text{step}$ (15.4k)	
the rows of the array of $\mathbf{a}:\mathbf{x}$ below its first are what step makes of $\mathbf{a}$ and the array of $\mathbf{x}$	
circuit	Hinze–Marsden
$\text{list}^+(A) \xrightarrow{A} \boxed{\text{newrows}} \rightarrow [[\text{tree}(A)]]$	
$= \text{list}^+(A) \xrightarrow{A} \boxed{\text{cons}} \rightarrow \text{list}^+(A) \xrightarrow{\text{init}} \boxed{\text{tail}} \rightarrow [\text{list}^+(A)] \xrightarrow{\text{list}(\text{row})} [[\text{tree}(A)]]$ definition of $\text{newrows} \triangleq \text{tic list}(\text{row})$, our name	
$= \text{list}^+(A) \xrightarrow{A} \boxed{\text{cons}} \rightarrow \text{list}^+(A) \xrightarrow{\text{init}} \boxed{\text{tail}} \rightarrow [\text{list}^+(A)] \xrightarrow{\text{list}(\text{mct}, \text{tail row}) \text{cons}} [[\text{tree}(A)]]$ (9.10) — (15.4g) — on the non-singleton tic lists	
$= \text{list}^+(A) \xrightarrow{A} \begin{cases} \boxed{\text{tops}} \\ \boxed{\text{rests}} \end{cases} \rightarrow \begin{cases} [\text{tree}(A)] \\ [[\text{tree}(A)]] \end{cases} \xrightarrow{\text{zip}} [\text{tree}(A) \times [\text{tree}(A)]] \xrightarrow{\text{list}(\text{cons})} [[\text{tree}(A)]]$ $\text{list}(f, g) = (\text{list}(f), \text{list}(g))$ zip; products; definitions of tops and rests	
$= \text{list}^+(A) \xrightarrow{A} \boxed{\text{array}} \xrightarrow{\text{step}} [[\text{tree}(A)]]$ (15.4i) and (15.4j); products; definition of step	

cons array=(1×array)addcol (15.4l) the array of a consed onto x is what addcol makes of a and the array of x	
circuit	Hinze–Marsden
$= \text{list}^+(A) \xrightarrow{\text{cons}} \text{list}^+(A) \xrightarrow{\text{inits}} [\text{list}^+(A)] \xrightarrow{\text{list}(\text{row})} [[\text{tree}(A)]]$ definition of array	
$= \text{cons } \text{inits} = (\pi_1 \text{ wrap}, \text{tic}) \text{ cons, abbreviating cons inits tail by tic}$	
$= \text{list}^+(A) \xrightarrow{\text{wrap}} \text{list}^+(A) \xrightarrow{\text{row}} [\text{tree}(A)] \xrightarrow{\text{cons}} [[\text{tree}(A)]]$ cons list(f)=(f×list(f)) cons; definition of newrows	
$= \text{list}^+(A) \xrightarrow{\text{newrows}} \text{list}^+(A) \xrightarrow{\text{tip}} \text{tree}(A) \xrightarrow{\text{wrap}} [\text{tree}(A)] \xrightarrow{\text{cons}} [[\text{tree}(A)]]$ wrap row=tip wrap	
$= \text{list}^+(A) \xrightarrow{\text{array}} \text{addcol} \xrightarrow{\text{addcol}} [[\text{tree}(A)]]$ (15.4k); products; definition of addcol	

§15.5. Data compression

(15.5a)

$[A] ::= \text{nil} \mid \text{snoc } ([A], A), \text{ list}^+ A ::= \text{wrap } A \mid \text{snoc } (\text{list}^+ A, A), \text{ String} = [\text{Char}], \text{ Code} ::= \text{sym Char} \mid \text{ptr} (\text{String}, \text{String}^+), F(\text{Code}, \text{String}) = 1 + (\text{String} \times \text{Code}), \alpha \triangleq [\text{nil}, \text{snoc}].$

$\text{decode} \triangleq ([\text{nil}, \text{extend}]) : [\text{Code}] \rightarrow \text{String}, \text{ extend } (xs, \text{sym } a) = xs \# [a], \text{ extend } (xs, \text{ptr } (ys, zs)) = xs \# zs \text{ when } ys \# zs \text{ is a proper prefix of } xs \# zs; H = \text{decode}^\circ.$

$\text{size} \triangleq ([\text{zero}, \text{distr } [1 \times c, 1 \times p] \text{ plus}]) \text{ with } c, p \text{ the constant costs of a symbol and a pointer; } R \triangleq \text{size} \leq \text{size}^\circ, Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau)), \text{ the two } \tau \text{ on symbols and on pointers.}$

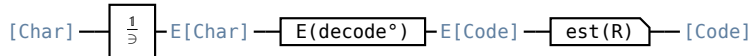
$\text{lrt } ws = \text{est}(\text{prefix}^\circ \times (\tau + \tau)) \{ (xs, (ys, zs)) \mid xs \# zs = ws, ys \# zs \text{ a proper prefix of } ws \}, \text{ the longest repeated tail; reduce } (ws \# [a]) = (zs \# [] \rightarrow [(ws, \text{sym } a), (xs, \text{ptr } (ys, zs))], [(ws, \text{sym } a)]) \text{ with } (xs, (ys, zs)) = \text{lrt } (ws \# [a]).$

$$0 \leq c \wedge 0 \leq p \Rightarrow (\mu X : \frac{[nil, extend]^\circ}{\exists} \text{thin}(Q) P([nil, (X \times 1) snoc]) est(R)) \sqsubseteq \frac{decode^\circ}{\exists} est(R), \quad (15.5b)$$

$$encode = (null \rightarrow nil, reduce \text{ list}((encode \times 1) snoc) \text{ minlist}(R))$$

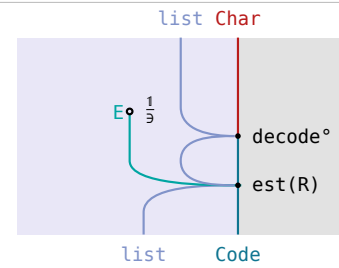
a smallest code sequence decoding to the given string is built from the right, each step emitting the last character as a symbol or ending with a pointer back into what has already been decoded

circuit



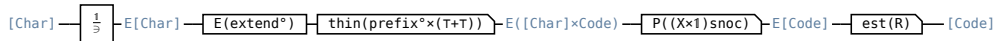
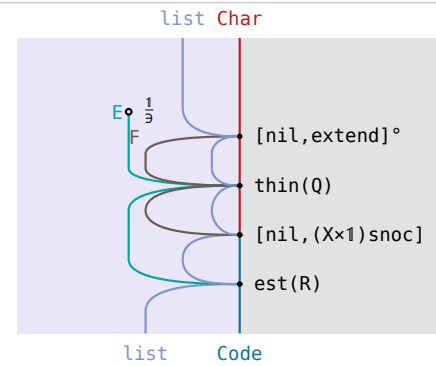
the specification $\dashv\vdash decode$

Hinze–Marsden

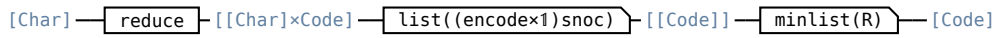
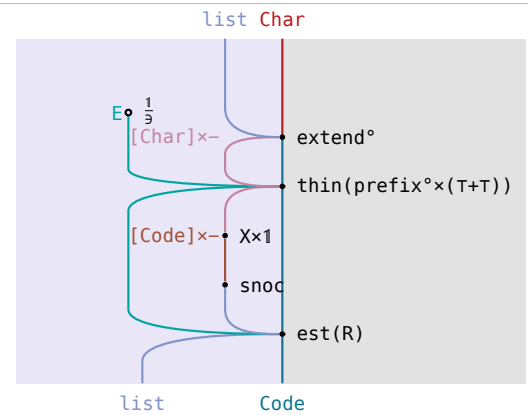


at $Q \triangleq F(\tau + \tau, prefix^\circ) = 1 + (prefix^\circ \times (\tau + \tau))$, the two τ on symbols and on pointers. Monotonicity $F(R) \alpha \sqsubseteq \alpha R$ is routine, the two costs being constants: $(\tau + \tau)[c, p] = [c, p]$.

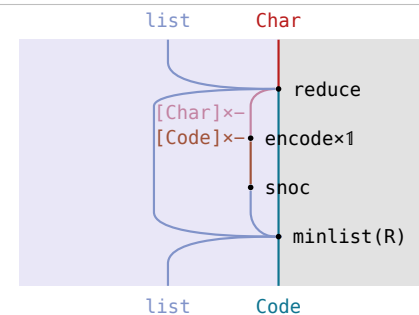
Proposition 9.4 splits the thinning condition in two — $F(\tau + \tau, R) \alpha \sqsubseteq \alpha R$, and $decode \text{ prefix} \sqsubseteq R^\circ \text{ decode}$, for which $decode \text{ init} \sqsubseteq R^\circ \text{ decode}$ is enough: dropping the last character of the output shortens or removes the last code element and never raises the cost. So between a symbol and a pointer nothing can be decided in advance, and between two pointers the longer match wins



Proposition 9.1: nil and $extend$ have disjoint ranges. The decompositions of one string are $\frac{extend^\circ}{\exists} (ws \# [a]) = \{(ws, \text{sym } a)\} \cup \{(xs, \text{ptr } (ys, zs)) \mid xs \# zs = ws \# [a], ys \# zs \text{ a proper prefix of } ws\}$ — take the last character as a symbol, or end with a pointer



$reduce$ implements $\frac{extend^\circ}{\exists} \text{thin}(prefix^\circ \times (\tau + \tau))$: thinning leaves at most two, the symbol and the pointer of the longest repeated tail lrt . Again exponential; the book gives no tabulation for it

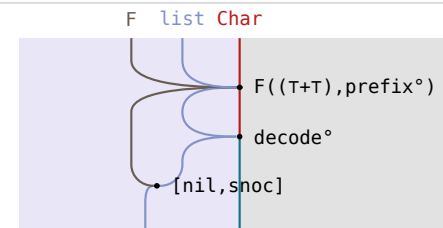
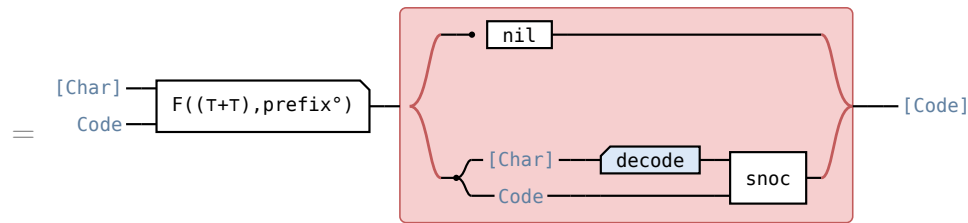
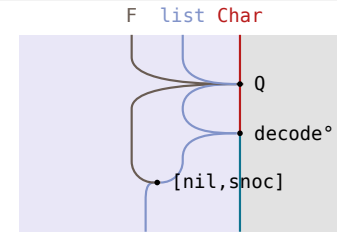
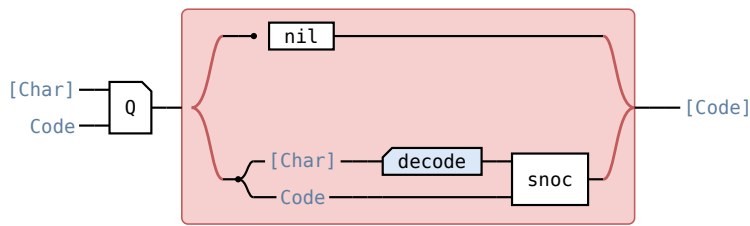


$$0 \leq c \wedge 0 \leq p \Rightarrow QF(\text{decode}^\circ)[\text{nil}, \text{snoc}] \sqsubseteq F(\text{decode}^\circ)[\text{nil}, \text{snoc}]R \quad (15.5c)$$

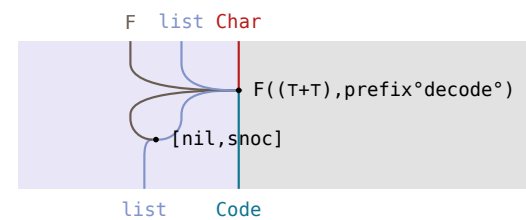
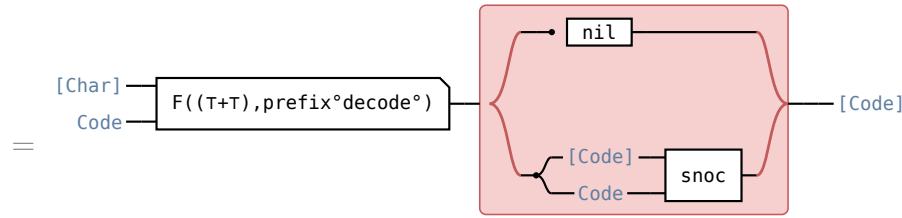
every code sequence built from a decomposition Q puts above a given one — a code element of the same kind, a longer front string — costs at least as much as one built from the given one

circuit

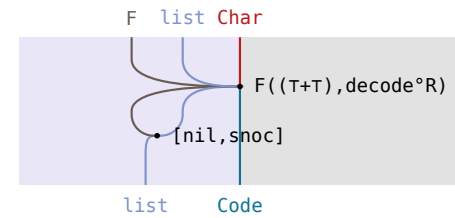
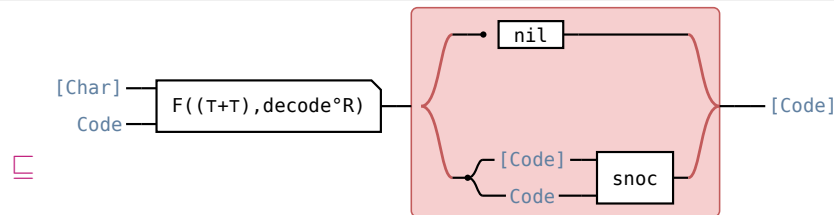
Hinze–Marsden



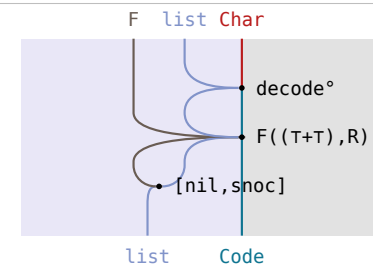
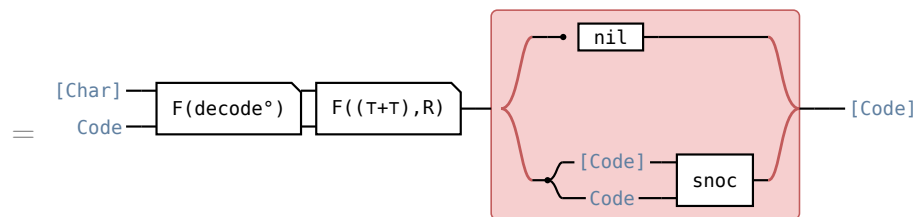
$Q \triangleq 1 + (\text{prefix}^\circ \times (T+T))$ is $F(T+T, \text{prefix}^\circ) \triangleq \text{decode}$



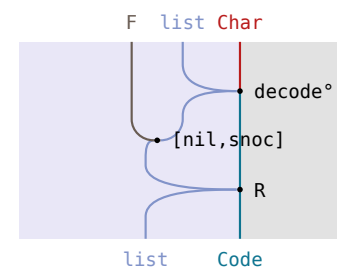
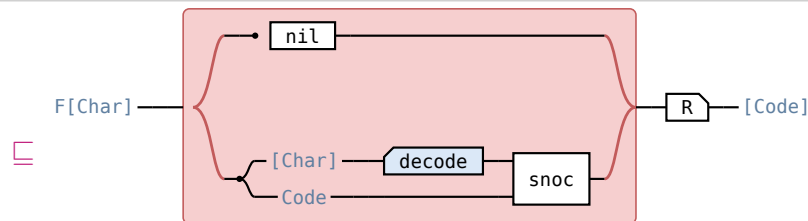
the bifunctor F preserves composition: $F(U, \text{prefix}^\circ)F(1, \text{decode}^\circ) = F(U, \text{prefix}^\circ \text{decode}^\circ)$



Proposition 9.4's second condition $\text{prefix}^\circ \text{decode}^\circ \sqsubseteq \text{decode}^\circ R$: dropping the last character drops the last **sym**, or shortens or drops the last pointer. Proved at **prefix** directly by induction, not through **init**



F preserves composition: $F(U, \text{decode}^\circ R) = F(1, \text{decode}^\circ)F(U, R)$



Proposition 9.4's first condition $F(T+T, R) \triangleq \alpha R$, left as an exercise in the book: **snoc** adds the cost of the last element to both sides, and $[c, p](T+T) = [c, p]$

extend nil^o=0 (15.5d) no string is both a result of extend and the result [] of nil	
circuit	Hinze-Marsden
nil returns only [], so $\text{nil}^o = \text{null } \text{nil}^o$	
$\text{extend } \text{null} = 0$: a symbol ends the string with a character, a pointer with its non-empty ZS	
0 composed with anything is 0	